18 (a) Express $\sqrt{3} + \sqrt{12}$ in the form $a\sqrt{3}$ where a is an integer.

(a) $\sqrt{12} = \sqrt{(4 \times 3)} = 2\sqrt{3}$

1 mark
Extract square factor 4.

$\sqrt{3} + 2\sqrt{3} = 3\sqrt{3}$

1 mark
$3\sqrt{3}$.

$3\sqrt{3}$

(2)

(b) Express $\left(\frac{1}{\sqrt{3}}\right)^7$ in the form $\frac{\sqrt{b}}{c}$ where b and c are integers.

(b) $(1/\sqrt{3})^7 = 1/(27\sqrt{3})$

1 mark
Simplify the seventh power.

Rationalise: $1/(27\sqrt{3}) \times \sqrt{3}/\sqrt{3}$

1 mark
Multiply by $\sqrt{3}/\sqrt{3}$.

$= \sqrt{3}/81$

1 mark
$\sqrt{3}/81$.

$\frac{\sqrt{3}}{81}$

(3)

(Total for Question 18 is 5 marks)